Overview of the math-proofs Collection

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Abstract

This document describes the result families in the `math-proofs` collection: machine-checked mathematics in which every proven result ships as a paper preprint (human-readable argument) paired with its Lean 4 formalization (machine-checked proof). Manuscripts are organized into families; each family groups a principal result with companion arguments, consequences, or alternative proofs. All formalizations typecheck with `lake build` under Lean 4.34.1 + Mathlib v4.34.1. The per-theorem verification status and axiom dependencies are recorded in `formalization.yaml` (machine-readable) and `FORMALIZATION_CATALOGUE.md` (human-readable).

1 Labeling convention

Label	Meaning
`[PROVED]`	Machine-checked by Lean; no axioms beyond Lean's standard ones.
`[PROVED*]`	Typechecks cleanly but depends on explicitly labeled classical-input axioms.
`[SOLVED]`	Computationally established on independent code paths (no Lean proof yet).

Table 1: Status labels used throughout the collection.

Nothing in this collection is claimed on the basis of a `sorry`.

2 Family 001 — Erdős–Straus identities and obstructions

Results on representing $4/n$ as a sum of three unit fractions.

ID	Result	Status	Lean module
ES-001	New covering identity for $n \equiv 217 \pmod{264}$	`[PROVED]`	`ESIdentity.lean`
ES-002	Finite-identity barrier on square classes mod 840	`[PROVED*]`	`ESBarrier.lean`

Table 2: Family 001 manuscripts.

ES-001 gives a new covering identity

$$\frac{4}{n} = \frac{1}{(n+3)/4} + \frac{1}{n(n+3)} + \frac{11}{n(n+3)}, \qquad n \equiv 217 \pmod{264},$$

covering the prime 1009 — the smallest prime missed by the published Mordell/Yamamoto/Rosati identities. The formalized theorem is `es_identity_217mod264`, depending only on Lean's standard axioms (`propext`, `Classical.choice`, `Quot.sound`).

ES-002 proves that no finite system of polynomial identities $4/(A_i t + B_i) = \sum 1/F_{ij}(t)$ with positive leading coefficients can cover a full square class mod 840: infinitely many primes escape every such system. Witness: $p = 77{,}598{,}049 \equiv 529 \pmod{840}$. The formalization carries two explicitly labeled classical-input axioms: `schinzel_mordell` and `dirichlet_arith_prog`.

3 Family 002 — Collatz convergent control (gap-factor theory)

A machine-checked theory of the gap factor $G(t)$ on from-above blocks of $\log_2 3$'s continued fraction, developed across thirteen Lean modules: the complete-quotient API for irrationals (`CollatzGapCF`), the C24 tail-mirror selector and strict dichotomy (`CollatzGapC24`, `CollatzGapC24Lemma3`), C23 sinh estimates (`CollatzGapC23`, `CollatzGapSinh`), the C22/C25/C26 assembly (`CollatzGapC22C25`), C18b convexity and the C26 margin-duality trichotomy (`CollatzGapConvex`), C29 discrepancy dominance and its from-below mirror (`CollatzGapC29`, `CollatzGapC29Mirror`), and the C64/C64b hump-law identity kernel (`CollatzGapHump`). All modules are `[PROVED]` with standard axioms only; preprints are in preparation.

4 Family 003 — Erdős–Straus witness-complex and QR-core theory

A machine-checked theory of Erdős–Straus covering families on refined moduli: the witness complex $W(M)$ (`WitnessComplexThm2`, `WitnessComplexThm13`), the T2 prime-square obstruction (`ESPrimeSquareT2`), the T1 two-case theorem (`EST1TwoCase`), and the QR-core — the set of quadratic-residue classes missed by both covering systems — with its exact count $|Q| = |S|/2^k$ (`ESQRCore`, `ESQRCoreCount`), the 5/7-forcing lemma (`ESFiveSevenForcing`), the mod-7 universal families and fiber-pinning lemma (`ESMod7UniversalFamilies`), the X0/S1S3 descent core (`ESX0S1S3`), general descent transfer (`ESDescentTransfer`) and the pair-specific conditional SDR endgame (`ESBranchSDREndgame`). All modules are `[PROVED]` with standard axioms only; preprints are in preparation.

5 Family 004 — Girth-5 extremal graphs

G5-001 is the 3v floor theorem for girth-5 graphs (`Girth5Floor3v.lean`, `[PROVED]`): the emitted-graph re-attachment model and cross-edge harmlessness argument, showing that no improving 3v move exists on the extremal graphs. Preprint in preparation.

6 Family 005 — q-adic classification of Erdős–Straus witnesses

QA-001 is the branch-1 q-adic type theorem (`ESQadicBranch1.lean`, `[PROVED]`): $v_q(r) \le v_q(A)$ on refined targets, the first classification theorem for the q-adic types of Erdős–Straus witnesses. Preprint in preparation.

7 Family 006 — Saturated Sidon sets

SS-001 (`[SOLVED]`): the set $\{16, 62, 66, 74, 88, 104, 117, 122, 123\}$ is Sidon and saturated in $[1, 184]$, verified on three independent code paths. Paper: `papers/r47-sat184-October-7-2026/`.

8 Family 007 — Erdős–Straus fiber compatibility and suppression

ES-004 ([PROVED]): the R185 fiber-compatibility kernel (sharp inverse-closure characterization for prime-power projections: `ESFiberCompat.lean`), its composite-modulus extension (`ESFiberCompatComposite.lean`), the fiber-$\{6, 11\}$ thinned-tiling catalogue (`ESFiberTiling.lean`), the composite suppression theory (suppression coefficient κ, persistence rate Π, the T4–T8 R-operator section: `CompositeSuppression.lean`), and the branch-2 degeneracy + divisor-closed family theorem (`ESBranchTwo.lean`, including the honest mod-8 refutation of the naive form). All five modules are machine-checked with standard axioms only. Paper: `papers/fiber-compatibility-composite-suppression-October-11-2026/`.

9 Family 008 — Sidon hole suppression and absorption

SS-002 ([PROVED]): R199/R200 hole-suppression finite-set lemmas and T5 saturation theorems (`HoleSuppression.lean`), the absorbed-regime arithmetic T5′ (`AbsorbedRegime.lean`), and the corrected T4 absorbed bound $c \leq 320$ (`AbsorbedBound.lean`). Paper: `papers/hole-suppression-absorption-Oc`

10 Family 009 — Translation marches and dip mechanics

SS-003 ([PROVED]): the surviving core of R227's translation march (rigid translation of blocked sets: `TranslationMarch.lean`), the corrected ψ-translate identities and chain/width bounds (`CorrectedMarch.lean`), the corrected unsaturated march with its additive bound (`TranslationMarchUnsat.lean`), the generalized hole-twin dip theorem (`HoleTwinDip.lean`), and the R247 dip-march sliding-window mechanics with φ-duality (`DipMarch.lean`). Paper: `papers/translation-march-dip-October-11-2026/`.

11 Family 010 — Sub-family dichotomy and ladder ceilings

SS-004: the R245 generalized sub-family theorem (`generalized_subfamily_identity` in `ESSubfamilyExtended.l` [PROVED]) with staged [PROVED*] extensions (R253 admission criterion, R261 A_0-constancy, R265 repaired split/layered), and the R259/R263 ladder-length ceiling (`theorem6`, `catalog_ceiling` in `MarchLadderCeiling.lean`, [PROVED]). Paper: `papers/subfamily-ladder-ceiling-October-11-2026/`.

12 Family 011 — Sigma transfer and fiber overlap

SS-005 ([PROVED]): R275 σ-transfer ascent-transfer machinery (`SigmaTrans.lean`) and the R277 fiber-overlap lemmas (`FiberOverlap.lean`). Paper: `papers/sigma-transfer-fiber-overlap-October-11-2026/`

13 Family 012 — Abstract locks and transition DAGs

SS-006 ([PROVED]): the repaired abstract lock theory (consecutive-firing characterization with the $W = \{0, 2\}$ sharpness witness: `AbstractLock.lean`) and the D7′ dual-ascent barrier plus L_s/C3 ascent-depth machinery (`TransDAG.lean`). Paper: `papers/abstract-lock-transdag-October-11-2026/`.

14 Family 013 — Void frames and defect calculus

SS-007: void frames under union forcing, V1–V8 (`VoidUnion.lean`, [PROVED]), void locks (`VoidLock.lean`, [PROVED]), the parameterized void-frame monoid N'_W (`VoidMonoidParam.lean`, [PROVED*] staged

verification, build-clean — a required build dependency), the D2(c) repair + D3 tiling (`VoidMonoid.lean`, `[PROVED]`), and the defect-homomorphism rigidity/counts (`VoidDefectHom.lean`, `[PROVED]`). Paper: `papers/void-frames-defect-October-11-2026/`.

15 Family 014 — ES absorption order and forcing spectrum

ES-005 (`[PROVED]`): the ES absorption poset and redundancy rank — A1–A3, ρ/R3′/R3″/T/R4′ superadditivity, singleton factorization (`ESAbsorptionOrder.lean`); the forcing spectrum partition S2(a)′/S2(b–d)/S3/S4 (`ESForcingSpectrum.lean`); and the forcing decomposition theorems A–D (`ESForcingDecomp.lean`). Paper: `papers/es-absorption-forcing-October-11-2026/`.

16 Family 015 — Sieve inversion and branch isolation

ES-006: the 44-CRT sieve inversion (`crt44_forward/backward/iff` in `ESRefinedSieveInverse.lean`, `[PROVED]` with narrower-than-standard axioms; three `native_decide` theorems `[PROVED*]`) and u-branch isolation (`ESUBranchIsolation.lean`, `[PROVED]`, standard axioms). Paper: `papers/sieve-inverse-branch-isolation-October-11-2026/`.

17 Summary of families

Family	Subject	Modules
001	Erdős–Straus identities and obstructions	2
002	Collatz convergent control (gap-factor theory)	13
003	Erdős–Straus witness-complex and QR-core theory	11
004	Girth-5 extremal graphs	1
005	q-adic classification of ES witnesses	1
006	Saturated Sidon sets	0 (computational)
007	ES fiber compatibility and suppression	5
008	Sidon hole suppression and absorption	3
009	Translation marches and dip mechanics	5
010	Sub-family dichotomy and ladder ceilings	2
011	Sigma transfer and fiber overlap	2
012	Abstract locks and transition DAGs	2
013	Void frames and defect calculus	5
014	ES absorption order and forcing spectrum	3
015	Sieve inversion and branch isolation	2

Table 3: The fifteen result families and their formalization footprint.

Versions and citations

Corrections and revisions are recorded as new versions in `CHANGELOG.md`, with previously released versions remaining accessible in git history. To cite an individual manuscript, use the BibTeX block in its `papers/<slug>/CITATION.bib`.